chapter, this means that the impulse response of the antichirp system is found by preforming a left-for-right flip of the chirp system's impulse response. Interesting, but what is it good for?

Consider how a radar system operates. A short burst of radio frequency energy is emitted from a directional antenna. Aircraft and other objects reflect some of this energy back to a radio receiver located next to the transmitter. Since radio waves travel at a constant rate, the elapsed time between the transmitted and received signals provides the distance to the target. This brings up the first requirement for the pulse: it needs to be as short as possible. For example, a 1 microsecond pulse provides a radio burst about 300 meters long. This means that the distance information we obtain with the system will have a resolution of about this same length. If we want better distance resolution, we need a shorter pulse.

The second requirement is obvious: if we want to detect objects farther away, you need more energy in your pulse. Unfortunately, *more energy* and *shorter pulse* are conflicting requirements. The electrical power needed to provide a pulse is equal to the energy of the pulse divided by the pulse length. Requiring both *more energy* and a *shorter pulse* makes electrical power handling a limiting factor in the system. The output stage of a radio transmitter can only handle so much power without destroying itself.

Chirp signals provide a way of breaking this limitation. Before the impulse reaches the final stage of the radio transmitter, it is passed through a chirp system. Instead of bouncing an impulse off the target aircraft, a chirp signal is used. After the chirp echo is received, the signal is passed through an antichirp system, restoring the signal to an impulse. This allows the portions of the system that measure distance to see short pulses, while the power handling circuits see long duration signals. This type of waveshaping is a fundamental part of modern radar systems.